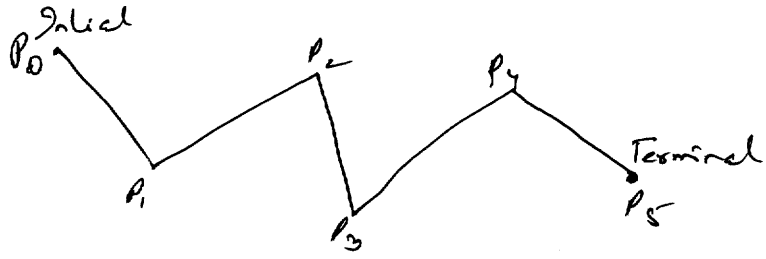
Geometric Forms and Models

One of the major concepts in computer Graphics is modelling of objects and pictures. By this we mean description of the objects and pictures for the computer so as to produce a visual display that simulates the real thing.

Simple Geometric Forms –

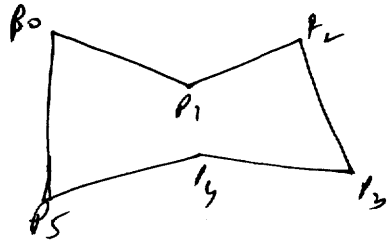
1. **Points and Lines** – are the basic building blocks of CG. We specify a point by giving its coordinates in 3 or 2 dimensional space. A line segment is specified by giving its endpoints $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$.

2. **Polylines** – A polyline is a chain of connected line segments. It is specified by giving the vertices $P_0, \ldots P_N$ defining the line segments. The first vertex is called the initial or starting point and the last vertex, the final or terminal point.



3. **Polygons** – A polygon is a closed polyline, i.e., one in which the initial and terminal points coincide. A polygon is specified by its vertex list $P_0, \ldots P_N, P_0$. The line segment $\overline{P_0P_1}, \overline{P_1P_2}, \ldots \overline{P_NP_0}$ are called edges of the polygon.

   A planar polygon is a polygon in which all vertices lie on the same plane.



Wireframe Model :-

A wireframe model consists of a collection of points (vertices) and a set of edges that connects pairs of points. Together the points and edges define the shape of the object. The vertices and edges are typically stored in databases that is accessed during the drawing process.

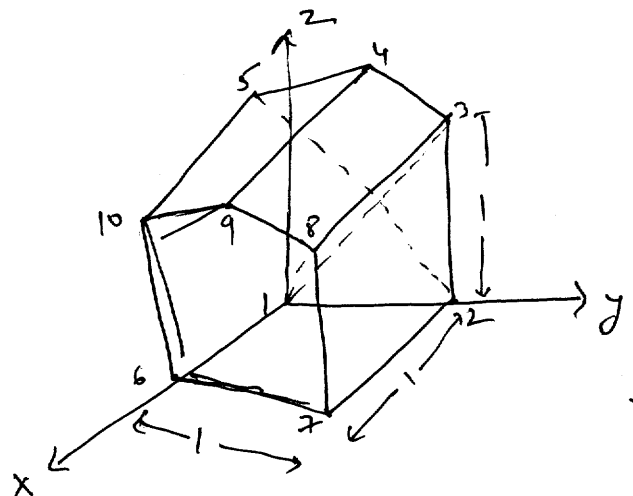
OR

A wireframe model consists of edges, vertices and polygons. Here vertices are connected by edges, and polygons are sequences of vertices or edges. The edges may be curved or straight lines. In the latter case, the wireframe model is called a polygonal net or polygonal mesh.

Drawing wireframe tends not be realistic: The objects appear hollow and can be hard to interpret. The use of shading and hidden surface removal tech. improve the conditions drastically, They are often used for previewing objects in an interactive scenario basically CAD/CAM. Under the wireframe model is the underlying solid object whose shape is suggested by the wireframe.

For both wireframe models and mathematical surfaces, the points and edges lie in 3D space, and so to render pictures of them something must be projected onto the drawing surface.

A wireframe model captures the shape of a 3D object in two lists, a vertex list and an edge list.



The basic barn

We examine the basic ideas of wireframe models using a polyhedron called basic barn.

The Vertex list specifies geometric information; where each corner is located. The vertices of the barn are defined here in a right handed coordinate system with z-axis upward, in accordance with modeling conventions.

Vertex List

Vertex	x	y	z	
1	0	0	0	
2	0	1	0	back side
3	0	1	1	
4	0	.5	1.5	
5	0	0	1	
6	1	0	0	
7	1	1	0	front side
8	1	1	1	
9	1	.5	1.5	
10	1	0	1	

The edge list provides connectivity information, specifying the two vertices that form the endpoints of each edge. Two extra edges are added to the backside to make it easier to distinguish the front and back of the barn, once drawing

Edge List

edge	vertex1	vertex2
1	1	2
2	2	3
3	3	4
4	4	5
5	5	1
6	6	7
7	7	8
8	8	9
9	9	10
10	10	6
11	1	6
12	2	7
13	3	8
14	4	9
15	5	10
16	2	5
17	1	3

Representing a Polygonal Net Model — There are several ways of representing a polygonal net model.

1. Explicit Vertex list $V = \{P_0, P_1, \ldots P_n\}$ - The points $P_i(x_i, y_i, z_i)$ are the vertices of the polygonal net, stored in the order in which they would be encountered by traveling around the model. When displaying model by drawing the edges, shared edges are drawn several times.

2. Polygon Listing — In this form of representation, each vertex is stored exactly once in a vertex list $V = (P_0, \ldots P_n)$ and each polygon is defined by pointing or indexing into this vertex list. Shared edges are drawn several times in displaying model.

3. Explicit Edge Listing :— In this form of representation, we keep a vertex list in which each vertex is stored exactly once and an edge list in which each edge is stored exactly once. Each edge in the edge list points to the 2 vertices in the vertex list which defines that edge. A polygon is now represented as a list of pointers or indices into the edge list. Additional information such as those polygons sharing a given edge, can also be stored in the edge list. Explicit edge listing can be used to represented the more general wireframe model. The wireframe model is displayed by drawing all the edges, and each edge is drawn only once.

Polyhedron = A polyhedron is a closed polygonal net (i.e one which encloses a definite volume) in which each polygon is planer. The polygons are called faces of the polyhedron In modeling, polyhedrons are quite often treated as solid objects, as opposed to wireframe.

Adv & Disadv. of Wireframe models :—

① used in engineering applications. Easy to construct and if they are composed of st. lines, easy to clip & manipulate through the use of geometric and coordinate transformations.

⇒ However for building realistic models, especially of highly curved objects we must use a very large number of polygons to achieve the illusion of roundness and smoothness.